Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	**Name**	Michael Grosshauser	**Email**	Michael.grosshauser@lbf.fraunhofer.de
Institute Name	Fraunhofer Institute for Structural Durability and System Reliability LBF			
Address	Schlossgartenstraße 6, 64289 Darmstadt, Germany			

2. Technology Overview

Title of Proposed Technology	*Tailor-made radical generators for FR synergy and reactive melt modification*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ), Others (√)			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Chemicals		*Chemicals*	
	Medium Category	Small Category	Medium Category	Small Category
	Fine Chemicals	*Additives*	*Polymeric Materials*	*Polymer Processing Technologies*
Concept of Proposed Technology	*Core concept: Design polymer-compatible, low-migration radical generators with tunable decomposition windows (Ton/T½ ≈ 180–240°C) that deliver dual functionality:* • *Flame-retardant synergy with phosphorus FRs (phosphinates/-ites/-ones), acting in gas and condensed phase to improve UL 94/LOI and reduce PHRR/smoke without antimony.* • *Reactive melt modification for virgin and recycled polymers (branching/coupling/grafting/IV rebuild), stabilizing MFR and mechanical properties over reprocessing cycles.* *Main functions:* • *Early-stage radical generation to enhance FR radical quenching (nitroxyl pathways) and char promotion; reduce total FR loading.* • *Melt repair: controlled chain extension/branching/coupling to recover rheology in PCR streams; reduce variability, VOC/odor, and tool fouling.* *Potential uses: E&E housings, cable ducts/connectors, automotive interiors/exteriors, appliance parts, building components (subject to local codes), and PCR upgrading lines (PP/PE/PA/PBT/PET/ABS).*			

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Rationale for Joint R&D Project	*Need: Industry seeks antimony-free, low-smoke FR systems that keep UL 94 performance while lowering total loading and stabilizing processing—especially for recyclates. Achieving controlled radical generation and robust FR synergy across diverse polymers requires mechanism-led design and pilot validation.* *Collaboration rationale:* • *Research institute brings additive concept design, structure–kinetics–performance modeling (ESR/TGA/DSC/FTIR/TDe-GC-MS), FR testing (UL 94/LOI/cone), melt rheology control, LCA/ESG evidence, and IP.* • *Korean SMEs contribute fine-chemical synthesis/scale-up, masterbatch/compounding, part fabrication, local conformity testing, and rapid path to domestic markets.* • *A joint effort de-risks scale-up, aligns to Korean test/certification routes, and accelerates commercialization in Advanced Mobility supply chains.*
Key Words	1. *Polymer-compatible radical generators* 2. *Sb-free FR synergy (with phosphorus FRs)* 3. *Reactive melt repair (branching/coupling/grafting)* 4. *Nitroxyl-generating HALS derivatives* 5. *Sulfonic/sulfinic ester radical systems*

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3-4)	End of the Project	Level (5-6)
R&D Objectives	• *N-alkoxy HALS families (NOR-type) with tailored substituents to set Ton/T½ in 180–240°C range; low volatility/migration; good solubility in PP/PA/PBT/ABS.* • *Halogen-free sulfonic/sulfinic acid esters (per WO2019020699) engineered as dual-use FR synergists and melt radical generators (dose-efficient, tunable kinetics).* • *Formulation modules with phosphorus FRs (Al/Zn phosphinates/-ites/-ones) optimized for PP, PA6/66, PBT, PC/ABS, HIPS/ABS.* • *Low-migration delivery: polymer-bound/oligomeric variants and masterbatch routes; optional microencapsulation for VOC/fog control.* • *Structure–kinetics–performance models (ESR/TGA/DSC/FTIR/TDe-GC-MS; Horowitz–Metzger analysis) linking additive chemistry, decomposition windows, FR effect, and melt repair.* • *Migration/VOC/fog screening and eco-tox/safety dossiers (REACH/K-REACH-ready).*			
R&D Target Specifications	*Flammability:* • *UL 94: PA6/66, PBT V-0 at 1.6 mm; PP V-2 at 1.6–3.2 mm with goal V-0 at 3.2 mm in selected grades.* • *LOI (ISO 4589-2): +3 vol% vs baseline; targets: ≥28% (PA/PBT), ≥23% (PP).* • *Cone calorimetry (ISO 5660): PHRR reduction ≥30% vs P-FR baseline; smoke SEA reduction ≥20%.* *Radical generator metrics:* • *Decomposition window Ton/T½ tunability within ±10°C of design target; minimal volatilization/blooming.* *Processing/recycling:* • *ΔMFR ≤10% after 3× re-extrusions at defined thermal/shear history; mechanical retention ≥90% vs baseline (E-modulus/impact) in target formulations.* *Emissions/migration:* • *VOC (e.g., VDA 278) ≤200 µg/g; Fog (DIN 75201-B) ≤3 mg; low migration (no visible blooming; extractables within product-specific limits after 7 days at 40°C).* *Loading efficiency:* • *Reduce total FR loading by ≥15–30% vs antimony-free reference while meeting target flammability ratings.*			
R&D Plan	*Approach* • *Mechanism-led additive design leveraging evidence that N-alkoxy HALS and related radical systems shift FR action earlier (gas/condensed phase), improving UL 94/LOI and cone metrics, and enabling controlled melt modification for PCR.* *Methodologies* • *Analytics: TGA/DSC (Ton/T½), ESR (radical generation), FTIR (bond evolution), TDe-GC-MS (volatiles mapping), Horowitz–Metzger kinetics.* • *Fire testing: UL 94 (IEC 60695-11-10), LOI/(N)OI where relevant, ISO 5660 cone calorimetry.* • *Processing/mechanics: compounding/injection/extrusion, MFR, DMTA/mechanical, reprocessing loops.* • *Emissions: VOC (VDA 278/ISO 16000), Fog (DIN 75201-B), migration/blooming.* • *Compliance: eco-tox and regulatory dossiers; LCA/ESG.* *Key tasks* • *T1: Molecule design & synthesis (HALS derivatives, sulfon/sulfinic esters, polymer-bound variants); safety data.*			

- *T2: Structure–kinetics screening and down-selection; masterbatch delivery forms.*
- *T3: FR synergy validation with P-FRs in PP/PA/PBT and styrenics; UL/LOI/cone tests.*
- *T4: Melt repair modules for PCR (branching/coupling/IV rebuild); processing window and variability reduction.*
- *T5: Scale-up with SMEs; application demos (E&E housings, automotive interior parts).*
- *T6: Certification testing and OEM-style validation; LCA/ESG; commercialization plan.*

Timeline and milestones (3 years)

Phase	**Months**	**Milestone**	**Description**
I	0–2	M1	Kick-off; detailed specs; compliance plan
I	0–6	M2	12+ prototype radical generators with characterized Ton/T½; initial UL/LOI/cone screens; shortlist per polymer
II	7–12	M3	Masterbatch scale-up; processing window maps; early VOC/fog/migration checks
II	13–18	M4	FR synergy proven (UL 94/LOI targets) in PP/PA/PBT; first melt repair success in PCR-PP (ΔMFR ≤10% after 3×)
II	19–24	M5	Application demo parts; cone calorimetry PHRR/SEA targets met
II	25–30	M6	Certification readiness; emissions/migration compliance; LCA/ESG report
II	31–36	M7	OEM-style validation; tech transfer package; IP/route-to-market plan

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• *Technical contributions: additive concept design; synthesis (bench scale); ESR/TGA/DSC/FTIR/TDe-GC-MS; FR testing (UL/LOI/cone); melt rheology and reprocessing; VOC/fog/migration protocols; LCA/ESG.* • *Research scope: PP, PA6/66, PBT, PC/ABS, HIPS/ABS; masterbatch strategies; structure–kinetics mapping; formulation optimization.* • *Resources: compounding/analytics/fire labs; project management; data packages for certification; IP generation.* • *Specific tasks: DOE, candidate down-selection, demo formulation authoring, documentation for Phase II submission.*
Expected R&R of Korean SME Partner	• *Technical contributions: fine-chemical synthesis and scale-up of HALS derivatives and sulfon/sulfinic esters; masterbatch/compounding; part fabrication.* • *Development tasks: processing window mapping; emissions/migration testing per domestic protocols; local conformity/certification coordination.* • *Testing/validation: run UL/IEC/ISO tests with Korean test organizations; assemble data for Phase I/II reviews; application validation in domestic supply chains.* • *Resource commitments: pilot equipment, materials, regulatory liaison (K-REACH); cost modeling; commercialization and tech transfer participation.*

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	• *Technological achievements: validated, tunable radical generator class delivering Sb-free FR synergy (UL 94/LOI, lower PHRR/smoke) and reactive melt repair for recyclates; low-migration delivery forms; robust structure–kinetics–performance models.* • *Commercialization potential: drop-in masterbatch products for Advanced Mobility parts; licensing/IP co-development; domestic scale-up with SMEs; certification-ready data packages.* • *Broader impact: reduced reliance on antimony/halogens; improved safety and ESG (lower smoke/odor/VOC); enhanced recyclate quality and consistency; competitiveness and export readiness for Korean SMEs in E&E/automotive.*

6. References

(1) Marney et al., Journal of Fire Sciences (2007): Mechanistic analysis of FR systems in PP; UL 94/LOI/(N)OI; TGA/FTIR/cone.

(2) Marney et al., Polymer Degradation & Stability (2008): Influence of N-alkoxy HALS on brominated phosphate FR; condensed/gas phase interactions; ESR/TDe-GC-MS.

(3) WO2019020699A1: Halogen-free sulfonic/sulfinic acid esters as FR synergists and radical generators.

(4) NOR-HALS patents (e.g., US 6,472,456; US 6,800,678; US 7,109,260): N-alkoxy HALS for synergistic FR action.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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